Explaining a classifier's decisions

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Abstract

As complex classifiers become increasingly prevalent in sensitive domains, the ability to understand and trust the decisions of these black-box models is essential. This poster provides a comprehensive overview of current explainable AI (XAI) methods designed to interpret classifier outputs. First, logic-based methods attempt to formalize the classifier's reasoning to offer formal guarantees on the quality and rigor of the produced explanations. Second, Shapley value methods evaluate feature importance by assigning a contribution score to each input based on game theory principles. Finally, perturbation methods observe model behavior by generating perturbed input samples. These methods provide local explanations either through linear approximations or user-friendly IF-THEN rules.

Introduction

Logic-based Abductive Explanations

We consider a classifier $\kappa : F \to Cl$ where F is the feature space and Cl the set of classes. An instance $x \in F$ is described by a set of literals (f_i, v). A partial assignment $E \subseteq x$ is a subset of literals of x, and $E(y)$ means that y agrees with E on all features it defines.

To define an Abductive explanation (AXp), we first need to define a weak AXp : Let $x \in F$ and $\kappa(x) = c$. A weak AXp of $\kappa(x)$ is a partial assignment $E \subseteq x$ such that:

$$\forall y \in F,\ E(y) \Rightarrow \kappa(y) = \kappa(x)$$

An AXp is a subset-minimal weak AXp. Intuitively, an AXp is the smallest set of feature values that is *sufficient* to guarantee the prediction, regardless of the values of the remaining features. There is multiple methods to Find AXp, we will describe three of them here that work with blackbox model.

CPI-Xp (Coverage-based Explanations)

Standard AXp's assume **feature independence**. When dependencies exist between features (e.g. a pregnant person is necessarily a woman), AXp's produce **superfluous or redundant** explanations [**cooper˙abductive˙2024**]. CPI-Xp solves this by incorporating **feature constraints** and selecting explanations with **maximal coverage**.

Definition. Let $F[\mathcal{C}]$ be the set of feasible instances under constraints $\mathcal{C}$, and $cov(E) = \{x \in F[\mathcal{C}] \mid E(x)\}$ the coverage of a partial assignment E. A **CPI-Xp** of $\kappa(x)$ is a partial assignment E such that:

1. $E(x)$
2. $\forall y \in F[\mathcal{C}],\ E(y) \Rightarrow \kappa(y) = \kappa(x)$
3. No valid explanation E' strictly subsumes E (i.e. $cov(E) \subsetneq cov(E')$)

The preferred variant **pCPI-Xp** keeps one representative per coverage class and is the only notion satisfying all seven formal properties (success, non-triviality, irreducibility, coherence, consistency, independence, non-equivalence).

Key characteristics:

- **Complexity:** Π_2^P-complete (exact), but a **polynomial approximation** exists via dataset-based versions (d-CPI-Xp) in $O(m^2n^2)$.

Efficient and Rigorous Model-Agnostic Explanations (sbAXp)

Classic logic-based XAI requires access to the model's internal structure and suffers from scalability issues. This method solves both by working **sample-based**: it only queries the model as a black-box on a finite sample $S \subseteq F$, producing **sample-based abductive explanations** (sbAXp).

Principle. The method exploits the **duality between AXp and CXp**: an AXp answers "Why this prediction?" while a Contrastive Explanation (CXp) answers "Why not another?". A CXp is a minimal subset of features that, if changed, suffices to alter the prediction. From all sbCXp's, the sbAXp's can be recovered via **minimal hitting sets**, and vice-versa.

Definition. A **sbAXp** is a subset $X \subseteq F$ such that:

$$\forall x \in S,\ \Big(\bigwedge_{i \in X} (x_i = v_i)\Big) \Rightarrow \kappa(x) = \kappa(v)$$

A **sbCXp** is a minimal subset X such that $\exists x \in S$ agreeing with v on $F \setminus X$ but $\kappa(x) \neq \kappa(v)$.

Key characteristics:

- **Complexity: Polynomial** $O(mn)$ to find one sbAXp — the most efficient among logic-based methods.
- **Smallest sbAXp:** Finding the *smallest* sbAXp is intractable in general, but solvable via a **MaxSAT** encoding in comparable time.
- **vs. CPI-Xp:** Much cheaper ($O(mn)$ vs. Π_2^P-complete), but lacks the 7 formal properties and does not handle feature dependencies.
- **vs. Argument-based:** Simpler algorithm (no argumentation framework needed), but does not filter contradictory explanations.

Argumentation-based Explanations

This method uses an **argumentation framework** to generate and refine sbAXp's. Instead of computing explanations directly, it builds **arguments** (plausible sbAXp's) and eliminates contradictions through **attacks**, producing more robust explanations.

Principle. Each argument has a **support** (a set of feature conditions) and a **conclusion** (a predicted class). Two arguments **attack** each other when their supports are compatible but their conclusions differ. All attacks are resolved simultaneously via **stable extensions** (maximal conflict-free sets). A selection function (e.g. Max, Card, $Incl_i$, $Incl_c$) filters the extensions, and an **inference rule** selects the final explanations:

- **Universal inference:** keeps only arguments present in *all* selected extensions — guarantees coherence.
- **Existential inference:** keeps arguments present in *any* selected extension — more explanations but may break coherence.

Key characteristics:

- **Complexity: Polynomial** — comparable to sbAXp.
- **Reliability:** ++ — highest among logic-based methods, thanks to contradiction elimination.
- **vs. sbAXp:** Same complexity but **filters out contradictory explanations**, producing more robust results.
- **vs. CPI-Xp:** Much cheaper and more robust (++), but does not provide the full 7 formal properties and does not handle feature constraints.
- **Trade-off:** Coherence is **only guaranteed with universal inference**; existential inference may produce contradictions.

1 Shapley values based XAI

This family of XAI methods are based on the Shapley values concept from game theory that give an **importance score** to each feature in the prediction of a single instance. Shapley values provide a mathematically fair and unique method to attribute the payoff of a cooperative game to the players.

Let $\mathcal{F} = \{1, \ldots, n\}$ be the set of the numbered features (the players). The Shapley value of a feature i is computed by going through all the coalitions not containing i and looking at the **payoff difference** with and without the feature i involved [**merrick˙explanation˙2020**] :

$$\phi_i = \frac{1}{n} \sum_{\mathcal{S} \in \mathcal{F} \setminus \{i\}} \binom{n-1}{|\mathcal{S}|}^{-1} \big(\mathbb{E}[f(X)|X_{\mathcal{S}\setminus\{i\}} = x_{\mathcal{S}\setminus\{i\}}] \tag{1}$$

$$- \mathbb{E}[f(X)|X_{\mathcal{S}} = x_{\mathcal{S}}]\big) \tag{2}$$

From the efficiency axiom of Shapley values [**fryer˙shapley˙2021**], we get the following property that states that the sum of the importance is equal to the actual predicted value minus a reference contribution that is the mean of all the predictions for the distribution on the input X

$$\sum_{i \in \mathcal{F}} \phi_i(v) = \mathbb{E}[f(X)|X_{\mathcal{F}} = x_{\mathcal{F}}] - \mathbb{E}[f(X)|X_{\emptyset} = x_{\emptyset}] \tag{3}$$

$$= f(x) - \mathbb{E}[f(X)] \tag{4}$$

Figure 1 vizualises computed importance scores for the prediction of a naive Bayes classifier on an instance of the Titanic dataset.

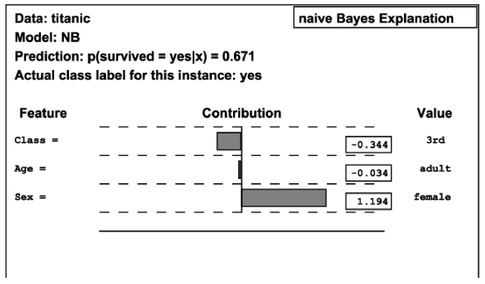

Figure 1: Feature importance attributions computed with Shapley values for the decision of a naive Bayes model on an instance of the Titanic dataset. From [**strumbelj˙efficient˙2010**]

Approximation

strumbelj˙efficient˙2010 have shown that computing exact Shapley values become intractable when the number of features is large ($n \geq 100$). **merrick˙explanation˙2020** have come up with sampling methods to get an approximation in polynomial time:

- sub-coalitions are sampled by taking all the features preceiding i in random orderings O_j of $\mathcal{F}$: $\mathcal{S}_j = Pre_i(O_j)$;
- the reference instances X are sampled following the distribution of input instances.

Perturbation model-agnostic method

Perturbation methods identify which input features most significantly influence a black-box model's decision process. These model-agnostic techniques compare the model's output for an original instance against its predictions for perturbed samples. By observing how these modifications affect the classification, we can quantify the importance of each feature without accessing the model's internal architecture.

LIME

LIME (Local Interpretable Model-Agnostic Explainer) generates local surrogates to approximate a black-box model f [**ribeiro˙why˙2016**].

Key Principle: It minimizes a loss function $\mathcal{L}(f, g, \pi_x)$ to ensure local fidelity while keeping the explanation model g simple enough ($\Omega(g)$) for humans. A most global explanation of the black-box model can be provided by SP-LIME which select a set of local and non-redundant LIME explanations.

Limitation: The explanation depends heavily on the kernel width π_x and sampling, leading to potential instability.

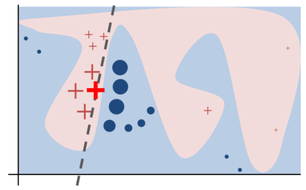

Figure 2: LIME linear local approximation. Black-box model f classes is represented in blue/gray. The red cross represents an explained instance and the others symbols are the perturbation samples generated from the original. The dotted line represents the local approximation of g.

Anchors

Anchors were developed to solve the lack of clear boundaries provided by LIME. To perform that, Anchors generates IF-THEN rules which 'anchor' a prediction. This set of short and disjoint rules are more comprehensible and then preferred by the users [**ribeiro˙anchors˙2018**].

Formal Definition: A rule A is an Anchor for x if $A(x) = 1$ and it maintains high precision τ among its neighbors $\mathbb{E}_{\mathcal{D}(z|A)}[1_{f(x)=f(z)}] \geqslant \tau$, with $\mathcal{D}(x \mid A)$ the distribution of neighbors of x which satisfy the Anchor A.

Computing Anchors is a combinatorial optimization problem. To handle the exponential search space, a **Bottom-Up approach** is used:

- **Candidate Generation**: Rules are extended iteratively with additional predicates.
- **Selection via MAB**: A Multi-Armed Bandit (*KL-LUCB*) algorithm identifies the most precise candidates using minimal model calls [**ribeiro˙anchors˙2018**].
- **Beam Search**: To avoid purely greedy results, the algorithm maintains B candidates, finally outputting the one with the **highest coverage**: $cov(A) = \mathbb{E}_{\mathcal{D}(z)}[A(z)]$.